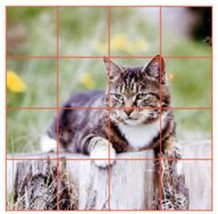
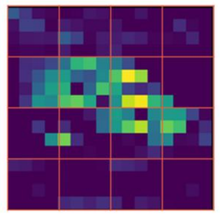
Differentially Private CutMix for Split Learning with Vision Transformer

Seungeun Oh, Jihong Park, Sihun Baek, Hyelin Nam, Praneeth Vepakomma, Ramesh Raskar, Medhi Bennis, and Seong-Lyun Kim

Motivation



(a) Input raw image.



(b) Smashed data of ViT.

(c) Smashed data of CNN.

- Recently, vision transformer (ViT) outpace CNN in CV field [1]. Split learning (SL) [2] with model partitioning is suitable for its large memory size and computing cost.

- **Objective**: to design privacy-preserving distributed ML considering differences between ViT and CNN.
  1. ViT often omits pooling: Efficient regularization on smashed data, but privacy leak.
  2. ViT learns global spatial information: Robust to large noise partially applied to the image (i.e. CutMix [3]).
  3. ViT runs at patch-level: Possibilities of a patch-scale regularizer.

  → Integrating the above yields a patch-level randomized CutMix of hidden representations (patch CutMix).

- To this end, we propose DP-CutMixSL, which combines patch CutMix with the differentially private (DP) SL framework.

References

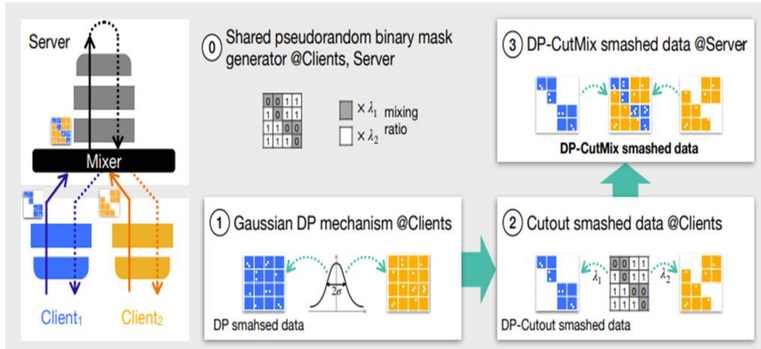
[1] A. Dosovitskiy, el al. *An image is worth 16x16 words: Transformers for image recognition at scale.* arXiv:2010.11929

[2] P. Vepakomma, et al. *Split learning for health: Distributed deep learning without sharing raw patient data.* arXiv:1812.00564

[3] S. Yun, et al. *Cutmix: Regularization strategy to train strong classifiers with localizable features.* IEEE/CVF ICCV 2019

[4] I. Mironov. *Rényi differential privacy.* IEEE CSF 2017

Proposed: DP-CutMixSL



Algorithm 1 DP-CutMixSL

requirements: $w = [w_{c,i}, w_s]^T$ ($w_{c,i}$: lower model segment, w_s: upper model segment)
η: learning rate
while w not converged **do**
  /*Runs on mixer*/
  samples $\{a_1, .., a_n\} \sim \mathrm{Dir}(\bar{\alpha})$
  generates pseudo random sequences M_i for all i ▷ *Pseudorandom binary mask generation*
  unicasts M_i to i-th client for all i

  /*Runs on client $i \in \mathbf{C}$***/**
  generates smashed data s_i by passing input data x_i through $w_{c,i}$
  produces $\bar{s}_i$ by masking s_i via M_i ▷ *Cutout smashed data*
  produces $\bar{s}'_i$ by applying Gaussian mechanism ▷ *DP-Cutout smashed data*
  uploads $\bar{s}'_i$ to the server

  /*Runs on server*/
  produces $\bar{s}'_i$ via $\bar{s}'_i$ aggregation for all i ▷ *DP-CutMix smashed data*
  generates loss $\sum_i L_i$ by passing $\bar{s}'_i$ through w_s in parallel
  updates w_s via $w_s \leftarrow w_s - \eta \cdot \nabla_{w_s}(\sum_i L_i)$ ▷ *Upper model segment update*
  unicasts i-th cut-layer gradient to i-th client for all i

  /*Runs on client $i \in \mathbf{C}$***/**
  updates $w_{c,i}$ via $w_{c,i} \leftarrow w_{c,i} - \eta \cdot \nabla_{w_{c,i}}(\sum_i L_i)$ ▷ *Lower model segment update*
end while

DP-CutMixSL's Operation:

1. Gaussian noise mechanism: Each device first injects random Gaussian noise into smashed data.

2. Random Cutout: Each device punches randomly selected patches of smashed data, yielding Cutout smashed data.

3. Random Mixup: These Cutout smashed data are uploaded to and put together by the server, resulting in DP-CutMix smashed data that continue FP.

Differential Privacy Analysis

Theorem 1. The RDP bounds [4] are given as:

$$\epsilon_o(\alpha) = \frac{\alpha}{2}\left(\frac{\Delta^2 D_x}{\sigma_x^2} + \frac{D_y}{\sigma_y^2}\right),$$

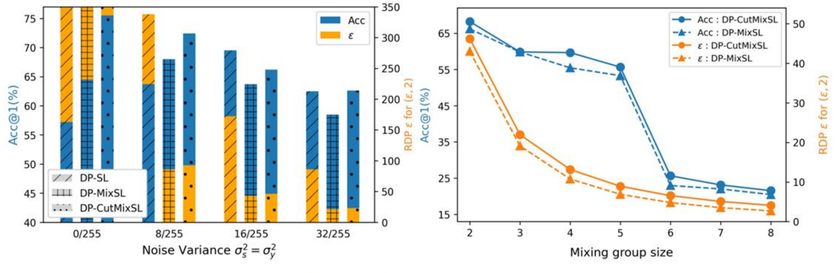
$$\epsilon_{Mix}(\alpha) = \frac{\alpha(\max_{i\in\mathbf{C}}\lambda_i)^2}{2}\left(\frac{\Delta^2 D_x}{\sigma_x^2} + \frac{D_y}{\sigma_y^2}\right),$$

$$\epsilon_{CutMix}(\alpha) = \frac{\alpha(\max_{i\in\mathbf{C}}\lambda_i)}{2}\left(\frac{\Delta^2 D_x}{\sigma_x^2} + \frac{D_y(\max_{i\in\mathbf{C}}\lambda_i)}{\sigma_y^2}\right).$$

1. Large privacy leakage in ViT: Without pooling, smashed data size D_x is large.
2. Cutouting & Mixing amplify DP: Mixup is the best for privacy, but accuracy. CutMix is a good compromise.
3. Hide in the crowd: Many clients inherently preserve privacy ("max"$_{i\in\mathbf{C}}\lambda_i$)

Numerical Evalution

Method ($\lvert\mathbf{C}=10\rvert$)	Models w/ CIFAR-10			Models w/ Fashion-MNIST		
	ViT-Tiny	PiT-Tiny	VGG-16	ViT-Tiny	PiT-Tiny	VGG-16
Standalone	48.84	47.77	54.97	77.65	78.21	80.12
SL [10]	57.21	52.28	62.62	85.68	82.35	84.39
SplitFed [14]	67.88	55.63	63.98	89.17	84.27	87.34
Standalone w. Cutout [20]	53.86	50.28	56.65	88.46	86.48	88.17
SL w. Mixup	69.23	64.89	**68.20**	88.21	87.62	88.53
SL w. Vanilla CutMix	71.78	58.21	33.50	87.86	86.31	89.01
CutMixSL (proposed)	**73.77**	**71.26**	67.53	**89.75**	**89.25**	**89.45**



1. CutMix SL achieves highst accuracy (Top)
2. Privacy-Accuracy trade-off, in terms of
   - Mixup vs. Cutout (Left)
   - Mixing Group Size vs. Accuracy (Right).